

2.10.43

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Question:

Let $\mathbf{a} = \mathbf{i} - \mathbf{k}$, $\mathbf{b} = x\mathbf{i} + \mathbf{j} + (1 - x)\mathbf{k}$ and $\mathbf{C} = y\mathbf{i} + x\mathbf{j} + (1 + x - y)\mathbf{k}$. Then $[\mathbf{a} \ \mathbf{b} \ \mathbf{c}]$ depends on

- ① only x
- ② only y
- ③ Neither x nor y
- ④ both x and y

Row Echelon form

$$[\mathbf{a} \ \mathbf{b} \ \mathbf{c}] = \begin{vmatrix} 1 & 0 & -1 \\ x & 1 & 1-x \\ y & x & 1+x-y \end{vmatrix} \quad (1)$$

$$R_2 \longrightarrow R_2 - xR_1 \implies \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ y & x & 1+x-y \end{pmatrix} \quad (2)$$

$$R_3 \longrightarrow R_3 - yR_1 \implies \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & x & 1+x \end{pmatrix} \quad (3)$$

$$R_3 \longrightarrow R_3 - xR_2 \implies \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \quad (4)$$

$$[\mathbf{a} \ \mathbf{b} \ \mathbf{c}] = (1) \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} - (0) \begin{vmatrix} 0 & 1 \\ 0 & 1 \end{vmatrix} + (-1) \begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix} = 1 \quad (5)$$

$$\implies [\mathbf{a} \ \mathbf{b} \ \mathbf{c}] = 1 \quad (6)$$

$\therefore [\mathbf{a} \ \mathbf{b} \ \mathbf{c}]$ is neither dependent on x and y .

```
// code5.c
double dot(double a[3], double b[3]) {
    return a[0]*b[0] + a[1]*b[1] + a[2]*b[2];
}

void cross(double u[3], double v[3], double res[3]) {
    res[0] = u[1]*v[2] - u[2]*v[1];
    res[1] = u[2]*v[0] - u[0]*v[2];
    res[2] = u[0]*v[1] - u[1]*v[0];
}
```

```
import ctypes

# Load the shared library (adjust filename & path if needed)
lib = ctypes.CDLL('./code5.so') # use 'vectorops.dll' on Windows

# Declare argument and return types for dot
lib.dot.argtypes = [ctypes.POINTER(ctypes.c_double), ctypes.
    POINTER(ctypes.c_double)]
lib.dot.restype = ctypes.c_double

# Declare argument types for cross
lib.cross.argtypes = [ctypes.POINTER(ctypes.c_double), ctypes.
    POINTER(ctypes.c_double), ctypes.POINTER(ctypes.c_double)]
lib.cross.restype = None
```

```
def dot(a, b):  
    a_c = (ctypes.c_double * 3)(*a)  
    b_c = (ctypes.c_double * 3)(*b)  
    return lib.dot(a_c, b_c)  
  
def cross(u, v):  
    u_c = (ctypes.c_double * 3)(*u)  
    v_c = (ctypes.c_double * 3)(*v)  
    res_c = (ctypes.c_double * 3)()  
    lib.cross(u_c, v_c, res_c)  
    return list(res_c)  
  
# Vector a = i - k  
a = [1, 0, -1]
```

Python Code

```
# Test values for x and y
xs = [0, 1, 2]
ys = [0, 1, -1]

print(x y [a b c])
for x in xs:
    for y in ys:
        b = [x, 1, 1 - x]
        c = [y, x, 1 + x - y]
        bc = cross(b, c)
        scalar_triple = dot(a, bc)
        print(f{x:<4} {y:<4} {scalar_triple:.2f})
```